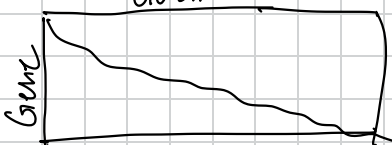


Protein Shape and Structure matters more than the actual sequences.

Finding Local Similarities:

With our current algorithm, the matches will be sparse throughout.



But you want



Local Alignment Problem

Input: Two strings v, w and a scoring matrix
 Output: Substrings of v, w whose best global alignment score is maximized over all substrings

Just add a zero-weight edge from the source to every node and the sink to every node.

Affine Alignment: Prefer alignments like $\begin{matrix} - & - & E \\ E & E & E \end{matrix}$ vs. $\begin{matrix} - & E & - \\ E & E & E \end{matrix}$

Affine penalty: gap opening penalty is higher than gap-extension penalty

The way we can do this efficiently is having a second dimension of size 3.

For multiple strings, you have multiple dimensions.
The runtime will blow up exponentially.

Fortunately, relatively simple heuristics go pretty far